ECON 0100 | Vignette C1 | Externalities

Solutions

Answers and work are shown in the red blocks.

Hogsmeade has recently seen a surge in shops selling Glittering Gum, a candy that makes one's words sparkle and glisten as one speaks. The Demand curve and Supply curve for Glittering Gum can be represented using the following relationships:

$$D: P_b = 30 - \frac{1}{2}Q_d \qquad S: P_s = 6 + \frac{1}{4}Q_s$$

After years of careful epidemiological analysis, a subcommittee of the Ministry tasked with improving the health and wellbeing of the wizarding community published a story in the Daily Profit establishing a link between the consumption of Glittering Gum and "Starlight Sneeze," a condition where non-consumers start sneezing out harmless but bothersome glitter. The estimate in the story is that this condition carries an inconvenience cost of 6 Coins per unit of Glittering Gum to those in Hogsmeade.

Use a graph and algebra to find equilibrium price, quantity, and DWL.

Equilibrium Price ($\hat{P}$): _____ Equilibrium Quantity ($\hat{Q}$): _____

Efficient Quantity (Q^*): _____ Deadweight Loss (DWL): _____

Equilibrium Price $\hat{P}$: 14 coins. Equilibrium Quantity $\hat{Q}$: 32 pieces of gum. Efficient Quantity Q^*: 24 pieces of gum. Deadweight Loss: 24 coins.

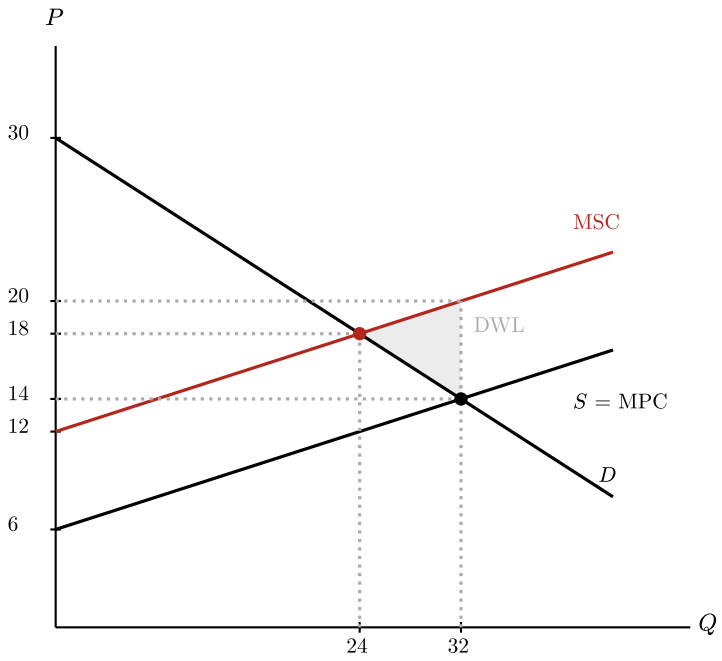

Figure 1: Glittering Gum market with the cost of the Starlight Sneeze

Every piece of gum chewed adds 6 coins of Starlight Sneeze on top of the cost of making it, so marginal social cost is supply shifted up by 6:

$$MSC : P = \left(6 + \tfrac{1}{4}Q\right) + 6 = 12 + \tfrac{1}{4}Q$$

Buyers and sellers do not pay for the sneezing, so the market settles where demand meets supply:

$$30 - \tfrac{1}{2}Q = 6 + \tfrac{1}{4}Q \quad \Rightarrow \quad \tfrac{3}{4}Q = 24 \quad \Rightarrow \quad Q = 32, \quad P = 30 - 16 = 14$$

The efficient quantity counts the sneezing, so it sits where demand meets marginal social cost:

$$30 - \tfrac{1}{2}Q = 12 + \tfrac{1}{4}Q \quad \Rightarrow \quad \tfrac{3}{4}Q = 18 \quad \Rightarrow \quad Q = 24, \quad P = 30 - 12 = 18$$

The market makes 8 pieces of gum too many. Each of the 25th through 32nd pieces is worth more to its buyer than it costs its seller, so it trades, but it is worth less than it costs society once the sneezing is counted. The 32nd, for example, is worth 14 to its buyer and costs society $12 + 8 = 20$. The losses on those pieces add up to the triangle between marginal social cost and demand from 24 to 32:

$$\text{DWL} = \tfrac{1}{2} \times (20 - 14) \times (32 - 24) = 24$$